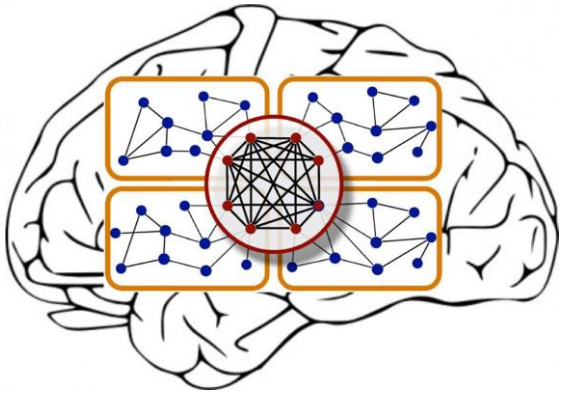


Anthony Sutton – MRes in Spatial Data Science and Visualisation

CASA, Bartlet, UCL - Dissertation 2018

New Horizons: Exploring London New Build Typology with Convolutional Neural Networks

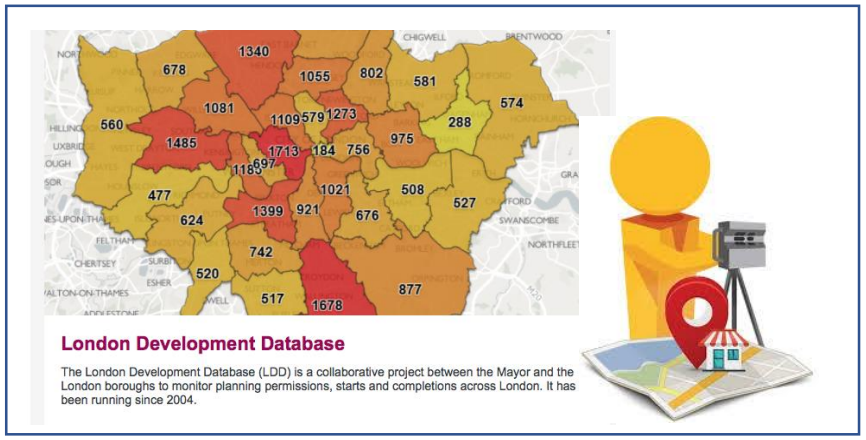
Aim

- Visual Exploration of New London
- Typological Analysis of London New Build

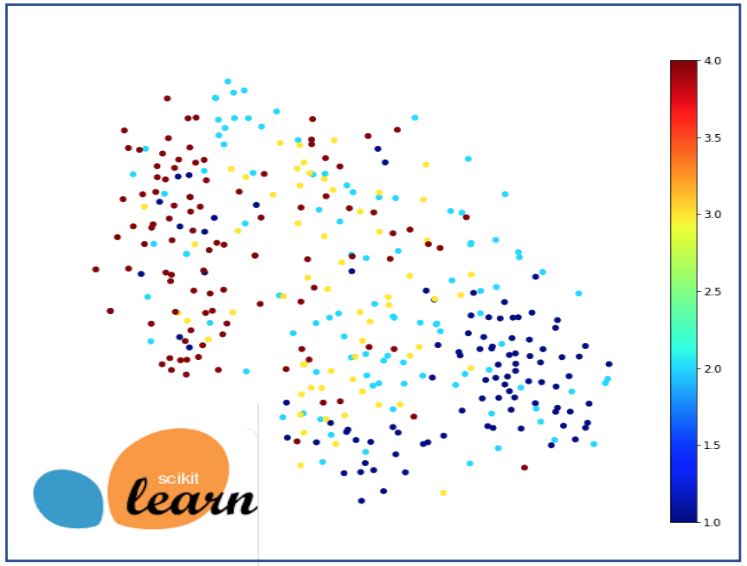
- CNN Image Classification Task on GLAs London Development Database

- Explore a Real-World Machine Learning Analysis Workflow

Method



- Data Model: London Development Database & Google Street View
  - CNN Transfer Learning with TensorFlow python
    - TSNE Sci Kit Learn Image Similar Analysis

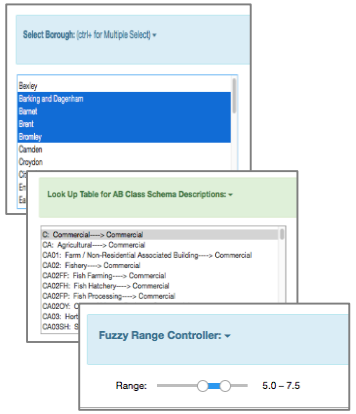


Machine Learning Workflow

MODEL BUILD

1 – EXPORE & BUILD DATA MODEL

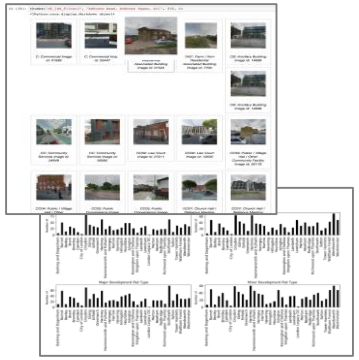
2 – BULK DOWNLOAD OF STREET LEVEL IMAGES



MODEL PREP

3 – PRE PROCESS IMAGES

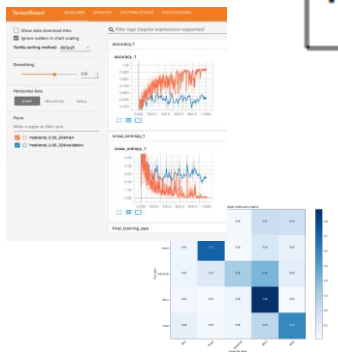
4 – LABEL SET CREATION



MODEL TRAIN

5 – CNN MODEL TRAINING

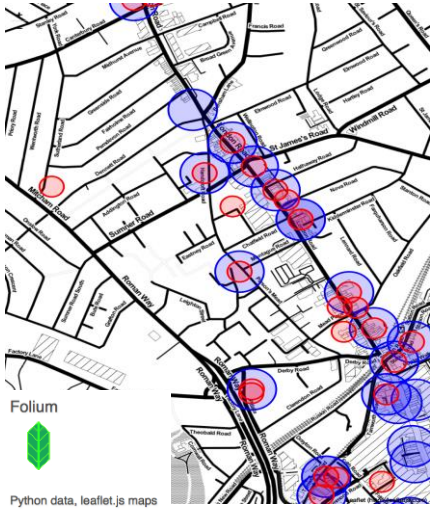
6 – LDD TEST SET CLASSIFICATION

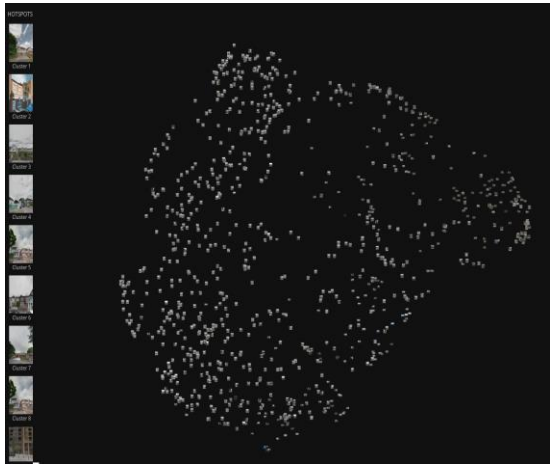


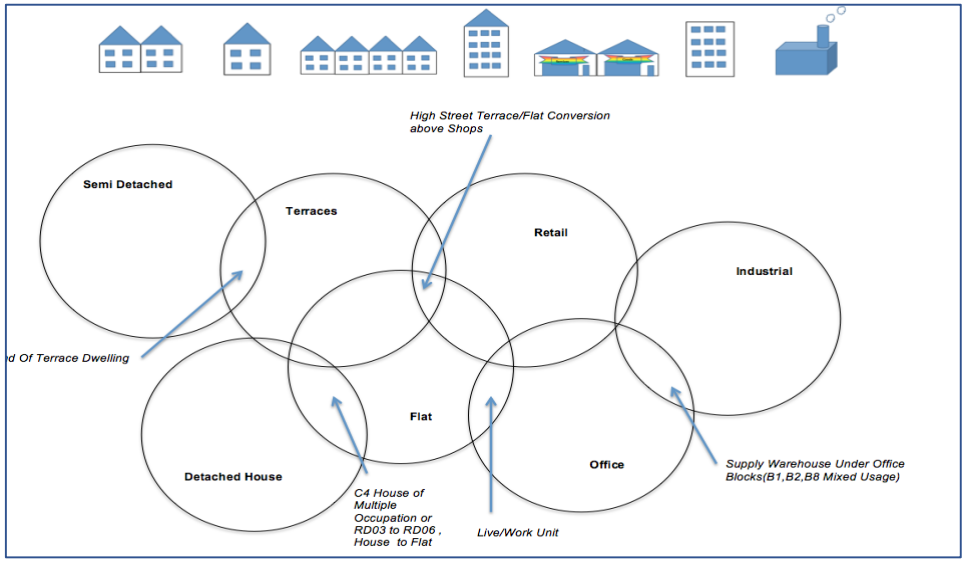
ANALYZE/VISUALIZE RESULTS

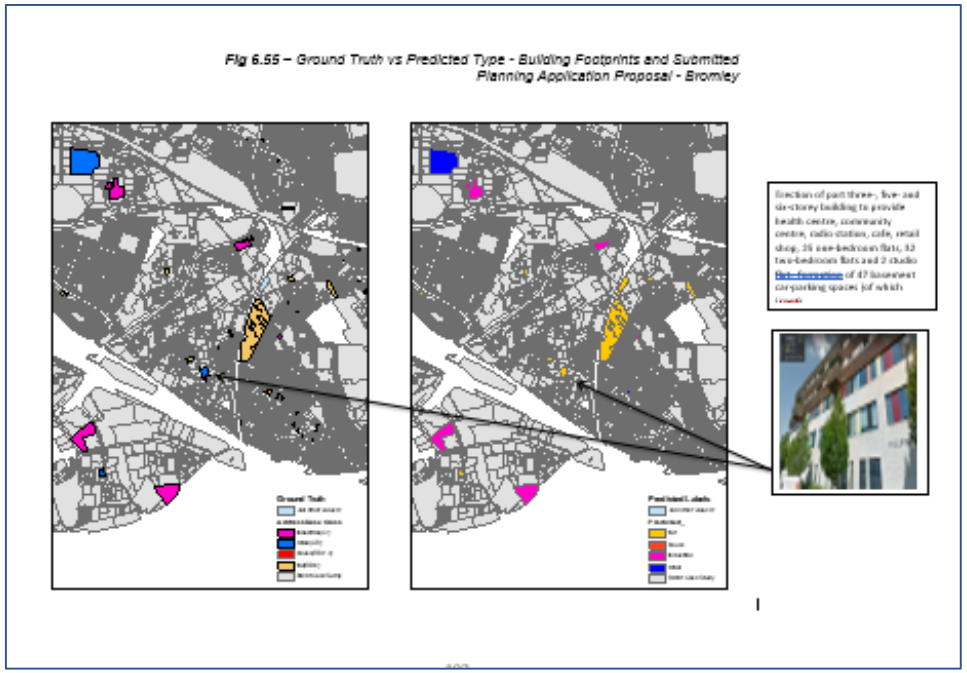
7 – SPATIAL ANALYSIS : BUILDING FOOTPRINT LOCATION(GROUND TRUTH MAPS)

8 – IMAGE SIMILAR ANALYSIS: TSNE VISUALISATION



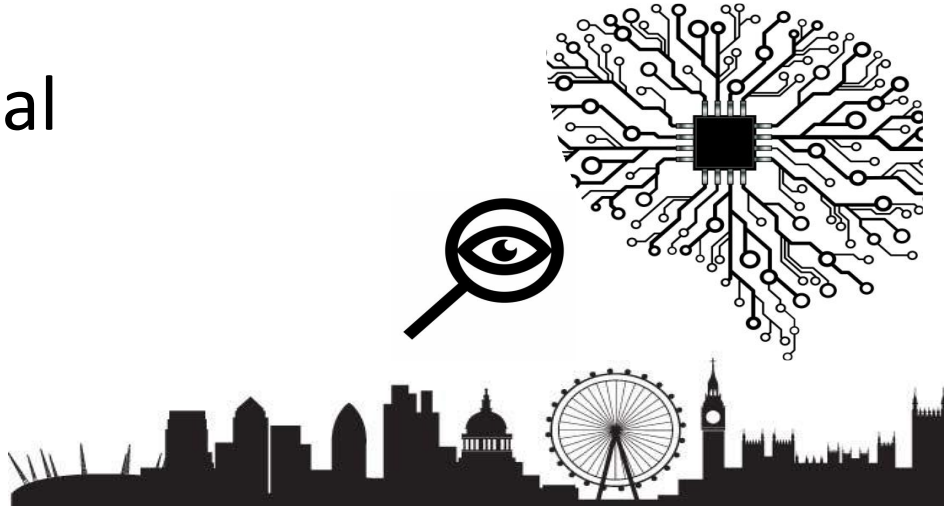






Analysis

- Model Overfit Contingency
  - Geo Reference Validation
  - CNN Fine Tuning
  - CNN Blackbox Visualization
- Predominant Types:
  - Mixed Use
  - High Density
  - Private Interface

Lessons Learned & Future Steps

- Cognitive Science: Elegant AI vs Hybrid ML
- Potential Applications: ML Assisted Mapping, Real Time Aggregate Urban Design & Property Mgmt. Systems
- Urban Visual Grammar and Gibson's Ecological Approach